

Riley Beenders

SOLUTIONS ARCHITECTURE | TECHNOLOGY STRATEGY | PLATFORM STANDARDIZATION

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PROFESSIONAL SUMMARY

Technology and engineering leader with 10+ years of progressive multidisciplinary experience, including CTO and Lead R&D roles. Establishes standardized architectures across hardware, software, and enterprise platforms, leads post-acquisition integration and standardization, and partners with vendors, global teams, and business stakeholders to deliver scalable, secure, and resilient systems.

CORE EXPERTISE

Architecture & Strategy: Solution and system architecture; platform standardization; technology strategy; architecture decisions balancing cost, reliability, and scalability; risk analysis and mitigations; emerging-technology evaluation (AI-assisted diagnostics, local LLM tooling)

Platforms & Integration: Kubernetes; Docker; Proxmox virtualization; Windows/Linux administration; ERP/CRM and Microsoft Business Central integration; TypeScript/React/Next.js (Node.js runtime); Python; Git/GitHub

Leadership & Delivery: Cross-functional and worldwide collaboration; hiring and technical interviews; vendor and third-party partner management; cost negotiation; requirements definition; technical documentation

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Re-engineered the POSEIDON and ICARUS product lines on a standardized architecture with swappable mounts, reducing support complexity and simplifying future scaling.
- Led post-acquisition IT integration for the division, standardizing software and internal processes with company-wide systems and implementing Microsoft Business Central ERP with SKUs synced across Proteor's global locations.
- Implemented ProteorPrint OS with an external software partner, enabling remote operation, AI-assisted diagnostics, and production metrics, and performed risk analysis to EU and U.S. standards across hardware and cloud products.
- Act as Hiring Manager for engineering roles, conducting technical interviews and owning final hiring decisions, and owned ICARUS-Lite through commercialization, contributing to increased revenue.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Progressed through four roles to CTO, setting technology strategy and overseeing the product-development pipeline and internal IT infrastructure.
- Designed and hosted a redundant Kubernetes cluster running a self-developed ERP/CRM that centralized orders, inventory, tasks, projects, and customer interactions, with dynamic BOM generation linked to purchasing.
- Integrated a modular multi-board CAN-bus control architecture and independently testable electronics enclosures, supporting larger, modular machines.
- Researched high-speed linear motors and early 6-axis robotic non-planar printing to inform next-generation platforms.

ADDITIONAL RELEVANT EXPERIENCE

RileyBeenders.com - Full-Stack Portfolio Project | 2026 | Built and deployed a TypeScript/React/Next.js site on Vercel with structured JSON content, a Node.js PDF endpoint, Git version control, a GitHub Actions workflow, and web analytics.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Built network infrastructure and deployed 750+ wireless access points across six school buildings; automated setup-data entry for 2,500+ Chromebooks with an Arduino-based tool.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: MathWorks Machine Learning and MATLAB Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)